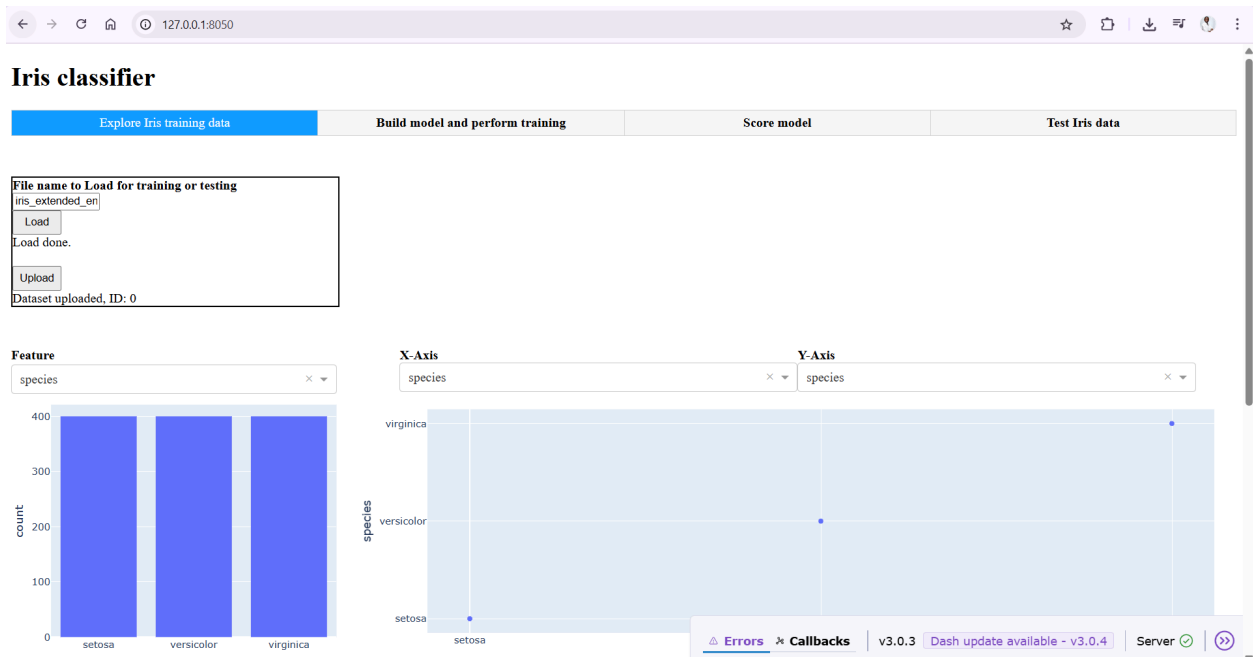


Install and test the “boto3” AWS SDK

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL FORIS python
Collecting s3transfer<0.13.0,>=0.12.0 (from boto3)
  Downloading s3transfer-0.12.0-py3-none-any.whl.metadata (1.7 kB)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in c:\users\student\thuotfl\lib\site-packages (from botocore<1.39.0,>=1.38.8->boto3) (2.9.0.post0)
Requirement already satisfied: urllib3<2.2.0,>=1.25.4 in c:\users\student\thuotfl\lib\site-packages (from botocore<1.39.0,>=1.38.8->boto3) (2.3.0)
Requirement already satisfied: six>=1.5 in c:\users\student\thuotfl\lib\site-packages (from python-dateutil<3.0.0,>=2.1->botocore<1.39.0,>=1.38.8->boto3) (1.17.0)
Downloading boto3-1.38.8-py3-none-any.whl (139 kB)
Downloading botocore-1.38.8-py3-none-any.whl (13.5 MB)
13.5/13.5 MB 670.3 kB/s eta 0:00:00
Downloading jmespath-1.0.1-py3-none-any.whl (20 kB)
Downloading s3transfer-0.12.0-py3-none-any.whl (84 kB)
Installing collected packages: jmespath, botocore, s3transfer, boto3
Successfully installed boto3-1.38.8 botocore-1.38.8 jmespath-1.0.1 s3transfer-0.12.0

[notice] A new release of pip is available: 25.0 -> 25.1.1
[notice] To update, run: python.exe -m pip install --upgrade pip
(thuotfl) PS C:\Users\STUDENT\OneDrive - andrew.cmu.edu\Documents\Spring2025\AISD\mini2lab4> python
Python 3.11.9 (tags/v3.11.9:de54cf5, Apr  2 2024, 18:12:12) [MSC v.1938 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> import boto3
>>> ddb = boto3.client('dynamodb')
File "<stdin>", line 1
    ddb = boto3.client('dynamodb')
    ^^^^^^^^^^^
SyntaxError: invalid syntax
>>> ddb = boto3.client('dynamodb')
>>> ddb.describe_limits()
{'AccountMaxReadCapacityUnits': 80000, 'AccountMaxWriteCapacityUnits': 80000, 'TableMaxReadCapacityUnits': 40000, 'TableMaxWriteCapacityUnits': 40000, 'ResponseMeta
data': {'RequestId': '773QAB95BJH5HE8IQ5HA2V4R9FVW4KQNS05AEMVJF66Q9ASUAAJG', 'HTTPStatusCode': 200, 'HTTPHeaders': {'server': 'Server', 'date': 'Sat, 03 May 2025 17
:37:03 GMT', 'content-type': 'application/x-amz-json-1.0', 'content-length': '143', 'connection': 'keep-alive', 'x-amzn-requestid': '773QAB95BJH5HE8IQ5HA2V4R9FVW4KQ
NS05AEMVJF66Q9ASUAAJG', 'x-amz-crc32': '3062975651'}, 'RetryAttempts': 0}}
```



Iris classifier

Explore Iris training data	Build model and perform training	Score model	Test Iris data
----------------------------	----------------------------------	-------------	----------------

Enter a dataset ID to use in training

0

New model

Model built and trained, ID: 0

Enter a model ID for re-training

Re-Train

Versicolor class

Iris classifier

Explore Iris training data	Build model and perform training	Score model	Test Iris data
----------------------------	----------------------------------	-------------	----------------

Enter a row text (CSV) to use in scoring

275.8,1.5,73.2,81.4,4.0,98.16,101300000000002,4

Enter a model ID for scoring

0

Score

Score result: Score done, class=1

Testing

The screenshot shows the AWS Lambda console for the function `IrisExtendedRetrainLambda`. A green banner at the top indicates "Successfully updated the function IrisExtendedRetrainLambda." Below this, the "Executing function: succeeded" status is shown with a link to logs. The "Details" section displays a JSON response: `{ "statusCode": 200, "body": "\\IrisExtendedRetrain Lambda return\\\" }`. A "Summary" table provides key metrics:

Code SHA-256	Execution time
0dJof9ZzIKOIUzh4nAsVDwkwCaVfwAKRtmK4yKdxCuQ=	11 seconds ago
Function version	Request ID
\$LATEST	fbf4b242-84d2-441d-a2eb-4c2d09bb5455
Duration	Billed duration
28.55 ms	29 ms
Resources configured	Max memory used
128 MB	80 MB

The "Log output" section shows the last 4 KB of the execution log, which includes a detailed JSON record of the function's execution, including the request ID, version, and various features and probabilities.

Created a test set with 1200 rows

The screenshot shows a Python script named `test_data.py` in a code editor. The script is designed to create a test dataset by sampling 200 rows from the `iris_extended_encoded.csv` file and altering 40 rows in the first column. The code includes comments explaining each step:

```
1 import pandas as pd
2 import random
3
4 # Load the dataset
5 df = pd.read_csv("iris_extended_encoded.csv")
6
7 # Sample 200 rows
8 sampled_df = df.sample(n=200, random_state=42).reset_index(drop=True)
9
10 # Determine the number of rows to alter (20% of 200 = 40)
11 num_to_alter = int(0.2 * len(sampled_df))
12
13 # Get the name of the first column
14 first_col = sampled_df.columns[0]
15
16 # Randomly pick 40 unique indices to alter
17 indices = random.sample(range(len(sampled_df)), num_to_alter)
18
19 # Shuffle the selected values and reassign to introduce interchanging
20 shuffled_values = sampled_df.loc[indices, first_col].sample(frac=1.0, random_state=99).values
21 sampled_df.loc[indices, first_col] = shuffled_values
22
23 # Save to CSV
24 sampled_df.to_csv("iris_extended_encoded_test.csv", index=False)
25
26 print(f"Generated iris_extended_encoded_test.csv with 200 samples and {num_to_alter} rows altered in the first column.")
27
```

The screenshot shows a Windows PowerShell terminal window with the following commands and output:

```
PS C:\Users\STUDENT> cd '.\OneDrive - andrew.cmu.edu\Documents\Spring2025\AISD\mini2lab4\'
PS C:\Users\STUDENT\OneDrive - andrew.cmu.edu\Documents\Spring2025\AISD\mini2lab4> python .\test_data.py
Generated iris_extended_encoded_test.csv with 200 samples and 40 rows altered in the first column.
PS C:\Users\STUDENT\OneDrive - andrew.cmu.edu\Documents\Spring2025\AISD\mini2lab4> |
```

Testing with a test set of 200 records randomly sampled and shuffled to introduce errors



IrisExtended Table Content

The screenshot shows the AWS DynamoDB console interface. The left sidebar contains navigation options: 'DynamoDB', 'Dashboard', 'Tables', 'Explore items', ' PartiQL editor', 'Backups', 'Exports to S3', 'Imports from S3', 'Integrations', 'Reserved capacity', and 'Settings'. The main area displays the 'IrisExtended' table with 50 items returned. The table has columns: 'partition_key (String)', 'sort_key (String)', 'Actual', 'Class', 'Features', and 'Probability'. The data is as follows:

	partition_key (String)	sort_key (String)	Actual	Class	Features	Probability
☐	18:45:38.812	abc	setosa	setosa	["elevation"...]	1.000
☐	19:17:52.557	abc	virginica	virginica	["elevation"...]	1.000
☐	18:48:04.626	abc	setosa	setosa	["elevation"...]	1.000
☐	18:44:17.759	abc	setosa	setosa	["elevation"...]	1.000
☐	18:46:53.444	abc	setosa	setosa	["elevation"...]	1.000
☐	18:44:44.860	abc	setosa	setosa	["elevation"...]	1.000
☐	18:47:53.730	abc	versicolor	versicolor	["elevation"...]	1.000
☐	18:47:24.647	abc	virginica	virginica	["elevation"...]	1.000
☐	18:46:52.923	abc	versicolor	versicolor	["elevation"...]	1.000
☐	18:47:20.224	abc	setosa	setosa	["elevation"...]	1.000
☐	18:46:47.456	abc	setosa	setosa	["elevation"...]	1.000
☐	18:46:40.831	abc	virginica	virginica	["elevation"...]	1.000

DynamoDB

Dashboard

Tables

Explore items

PartiQL editor

Backups

Exports to S3

Imports from S3

Integrations

Reserved capacity

Settings

DAX

Clusters

Subnet groups

Parameter groups

Events

Table: IrisExtended - Items returned (50)

Scan started on May 03, 2025, 21:23:10

	partition_key (String)	sort_key (String)	Actual	Class	Features	Probability
☐	19:17:59.543	abc	setosa	versicolor	["elevation"...	1.000
☐	19:17:52.557	abc	virginica	virginica	["elevation"...	1.000
☐	19:17:38.127	abc	setosa	setosa	["elevation"...	1.000
☐	18:48:04.626	abc	setosa	setosa	["elevation"...	1.000
☐	18:47:53.730	abc	versicolor	versicolor	["elevation"...	1.000
☐	18:47:36.253	abc	setosa	setosa	["elevation"...	1.000
☐	18:47:27.933	abc	versicolor	versicolor	["elevation"...	1.000
☐	18:47:24.647	abc	virginica	virginica	["elevation"...	1.000
☐	18:47:20.224	abc	setosa	setosa	["elevation"...	1.000
☐	18:47:04.338	abc	versicolor	versicolor	["elevation"...	1.000
☐	18:46:53.444	abc	setosa	setosa	["elevation"...	1.000
☐	18:46:52.923	abc	versicolor	versicolor	["elevation"...	1.000
☐	18:46:47.456	abc	setosa	setosa	["elevation"...	1.000

CloudShell

Feedback

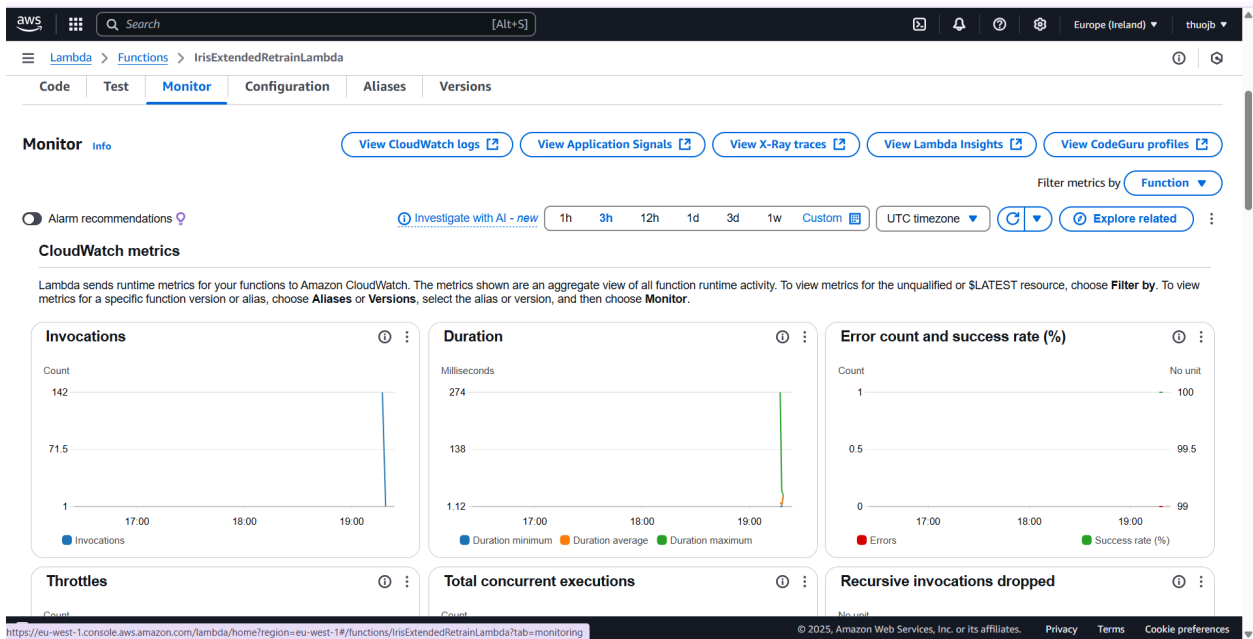
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Cloud Watch



eu-west-1.console.aws.amazon.com/lambda/home?region=eu-west-1#/functions/IrisExtendedRetrainLambda?newFunction=true&tab=monitoring

CloudWatch Logs

Lambda logs all requests handled by your function and automatically stores logs generated by your code through Amazon CloudWatch Logs. To validate your code, instrument it with custom logging statements. The following tables list the most recent and most expensive function invocations across all function activity. To view logs for a specific function version or alias, visit the **Monitor** section at that level.

Recent invocations

#	Timestamp	RequestId	LogStream	DurationInMS	BilledDurationInMS	MemorySetInMB	MemoryUsedInMB
1	2025-05-03T19:19:26.610Z	fbf4b242-84d2-441d-a2eb-4c2d099bb5455	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	28.55	29.0	128.0	88.0
2	2025-05-03T19:18:14.931Z	f89a3b8c-f0e7-4101-8f2d-ecbc474ccae5	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	1.53	2.0	128.0	88.0
3	2025-05-03T19:18:14.930Z	d013526a-9788-4bb6-9e65-64a719a53dfa	2025/05/03/[SLATEST]01a45cc8536b465f93f00b5b490735a0	1.8	2.0	128.0	88.0
4	2025-05-03T19:18:14.678Z	e6c5706e-7685-42b0-8c86-b48e99d494bb	2025/05/03/[SLATEST]01a45cc8536b465f93f00b5b490735a0	1.34	2.0	128.0	88.0
5	2025-05-03T19:18:14.177Z	246ae123-3c10-4d63-9509-2e77826de3d3	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	1.6	2.0	128.0	88.0
6	2025-05-03T19:18:14.177Z	d8b41f6b-6c61-47e5-a278-ee4a7d8cbfcd	2025/05/03/[SLATEST]01a45cc8536b465f93f00b5b490735a0	1.2	2.0	128.0	88.0
7	2025-05-03T19:18:13.679Z	6040715d-efaf-416c-838d-954d09024035	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	1.44	2.0	128.0	88.0
8	2025-05-03T19:18:13.428Z	2aba7fad-0bb7-47a4-b164-30a30ef4bf5	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	1.28	2.0	128.0	88.0
9	2025-05-03T19:18:12.931Z	d8494c3a-6b17-4347-8661-e8ae2cf7f576	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	6.31	7.0	128.0	88.0

Most expensive invocations in GB-seconds (memory assigned * billed duration)

#	Timestamp	RequestId	LogStream	BilledDurationInMS	MemorySetInMB	BilledDurationInGBSec
1	2025-05-03T19:17:36.328Z	6c3aa02a-c40e-48fd-bddf-2d6418c51fcb	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	243.0	128	0.030375
2	2025-05-03T19:17:27.551Z	44a621f4-7c86-4284-8314-46df693457ef	2025/05/03/[SLATEST]01a45cc8536b465f93f00b5b490735a0	231.0	128	0.028875
3	2025-05-03T19:17:36.388Z	047e92b4-6f84-4f97-9f36-98d4894be494	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	45.0	128	0.005625
4	2025-05-03T19:17:27.111Z	50447d79-9f7b-474d-8ecc-94c951e7bc0d	2025/05/03/[SLATEST]01a45cc8536b465f93f00b5b490735a0	44.0	128	0.0055
5	2025-05-03T19:18:06.708Z	24c403f2-1bdf-4d79-902b-d7c44b37d490	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	39.0	128	0.004875
6	2025-05-03T19:17:57.191Z	d0228188-c1c9-4124-a9be-05e72cb38ad8	2025/05/03/[SLATEST]01a45cc8536b465f93f00b5b490735a0	38.0	128	0.00475
7	2025-05-03T19:17:49.431Z	4c6f775a-8f49-499e-8ec5-84821e34358a	2025/05/03/[SLATEST]01a45cc8536b465f93f00b5b490735a0	37.0	128	0.004625
8	2025-05-03T19:18:01.448Z	10949588-23f3-40e6-aa4c-6a47fd5af28d	2025/05/03/[SLATEST]167551c1f8e84c018c2ddd01c90a37b0	36.0	128	0.0045

https://eu-west-1.console.aws.amazon.com/lambda/home?region=eu-west-1#/functions/IrisExtendedRetrainLambda?tab=monitoring

IrisExtendedRetrain Content

eu-west-1.console.aws.amazon.com/dynamodbv2/home?region=eu-west-1#item-explorer?maximize=true&table=IrisExtendedRetrain

Completed · Items returned: 33 · Items scanned: 33 · Efficiency: 100% · RCUs consumed: 4

Table: IrisExtendedRetrain - Items returned (33)

Scan started on May 03, 2025, 21:21:58

	partition_key (String)	sort_key (String)	Actual	Class	Features	Probability
☐	19:17:59.543	abc	setosa	versicolor	{"elevation":...	1.000
☐	19:17:44.563	abc	setosa	virginica	{"elevation":...	1.000
☐	19:17:27.731	abc	virginica	setosa	{"elevation":...	1.000
☐	19:17:34.902	abc	virginica	setosa	{"elevation":...	1.000
☐	19:17:36.514	abc	virginica	setosa	{"elevation":...	1.000
☐	19:17:52.003	abc	setosa	virginica	{"elevation":...	1.000
☐	19:18:07.138	abc	versicolor	virginica	{"elevation":...	1.000
☐	19:18:03.059	abc	virginica	setosa	{"elevation":...	1.000
☐	19:18:05.354	abc	versicolor	virginica	{"elevation":...	1.000
☐	19:17:49.313	abc	versicolor	versicolor	{"elevation":...	0.826
☐	19:17:57.343	abc	setosa	versicolor	{"elevation":...	1.000

CloudShell Feedback

eu-west-1.console.aws.amazon.com/dynamodbv2/home?region=eu-west-1#item-explorer?maximize=true&table=IrisExtendedRetrain

Search [Alt+S]

DynamoDB > Explore items > IrisExtendedRetrain

Table: IrisExtendedRetrain - Items returned (33)

Scan started on May 03, 2025, 21:32:51

< 1 > ⚙️

☐	partition_key (String)	sort_key (String)	Actual	Class	Features	Probability
☐	12:34:56.789	abc	versicolor	setosa	{"sepal_len..."}	0.85
☐	19:17:22.925	abc	setosa	virginica	{"elevation"..."}	1.000
☐	19:17:27.731	abc	virginica	setosa	{"elevation"..."}	1.000
☐	19:17:28.007	abc	virginica	setosa	{"elevation"..."}	1.000
☐	19:17:34.622	abc	versicolor	virginica	{"elevation"..."}	1.000
☐	19:17:34.902	abc	virginica	setosa	{"elevation"..."}	1.000
☐	19:17:36.241	abc	versicolor	setosa	{"elevation"..."}	1.000
☐	19:17:36.514	abc	virginica	setosa	{"elevation"..."}	1.000
☐	19:17:37.343	abc	setosa	versicolor	{"elevation"..."}	1.000
☐	19:17:43.519	abc	virginica	setosa	{"elevation"..."}	1.000
☐	19:17:44.563	abc	setosa	virginica	{"elevation"..."}	1.000
☐	19:17:49.313	abc	versicolor	versicolor	{"elevation"..."}	0.826

https://eu-west-1.console.aws.amazon.com/console/home?region=eu-west-1

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Clean up DynamoDB Tables

eu-west-1.console.aws.amazon.com/dynamodbv2/home?region=eu-west-1#tables

Search [Alt+S]

DynamoDB > Tables

Tables (2/2) Info

Find tables Any tag key Any tag value < 1 > ⚙️

☒	Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Rea
☒	IrisExtended	Active	partition_key (S)	sort_key (S)	0	0	Off	☆	On-
☒	IrisExtendedRetrain	Active	partition_key (S)	sort_key (S)	0	0	Off	☆	On-

CloudShell Feedback

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eu-west-1.console.aws.amazon.com/dynamodbv2/home?region=eu-west-1#tables

DynamoDB > Tables

Dashboard
Tables
Explore items
 PartiQL editor
 Backups
 Exports to S3
 Imports from S3
 Integrations **New**
 Reserved capacity
 Settings

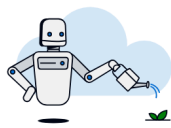
▼ **DAX**
Clusters
Subnet groups
Parameter groups
Events

✓ The request to delete the "IrisExtended, IrisExtendedRetrain" tables has been submitted successfully.

Tables (0) [Info](#)

Find tables Any tag key Any tag value < 1 > ⚙️

■ Name ▲ Status ▼ Partition key ▼ Sort key ▼ Indexes ▼ Replication Regions ▼ Deletion protection ▼ Favorite ▼ Read capacity n



You have no tables in this account in this AWS Region.

[Create table](#)

Lambda Function

eu-west-1.console.aws.amazon.com/lambda/home?region=eu-west-1#/functions

Lambda > Functions

Functions (1/1) Last fetched 10 seconds ago ⌂ Actions ▼ Create function

Filter by attributes or search by keyword

Function name	Description	Package type	Runtime	Last modified
IrisExtendedRetrainLambda				41 minutes ago

Delete function ✕

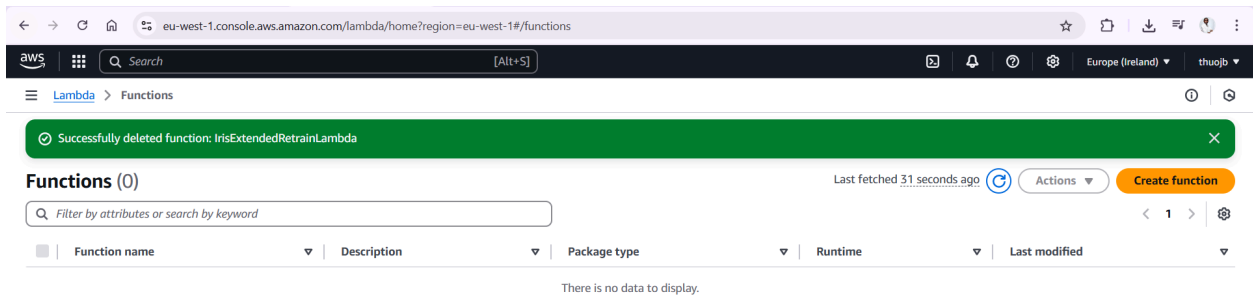
Delete function **IrisExtendedRetrainLambda** permanently? This action can't be undone.

⚠️ Deleting a function permanently removes the function code. The related logs, roles, test event schemas, and triggers are retained in your account.

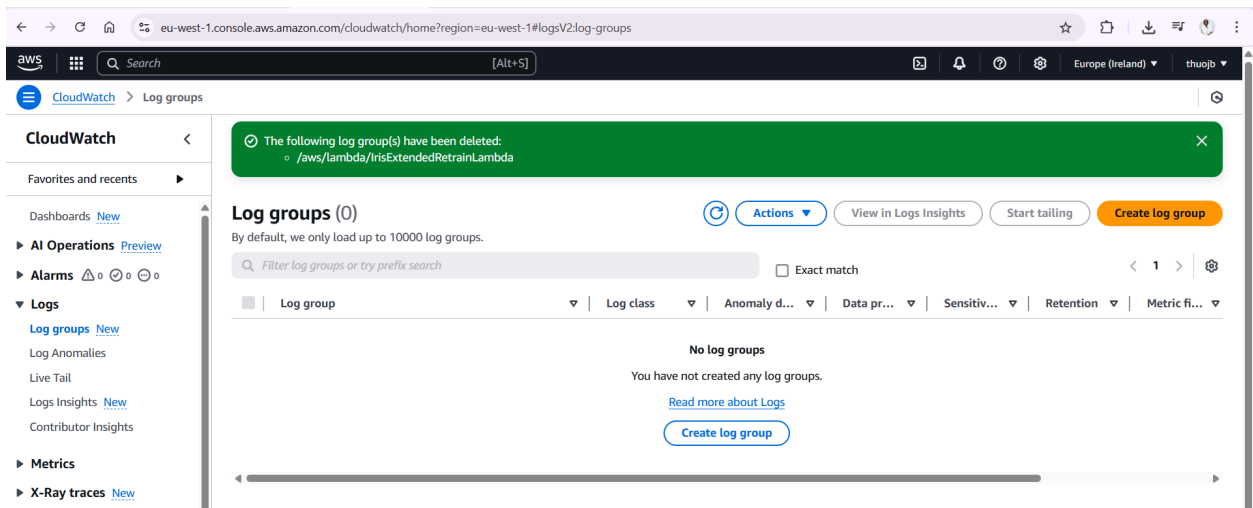
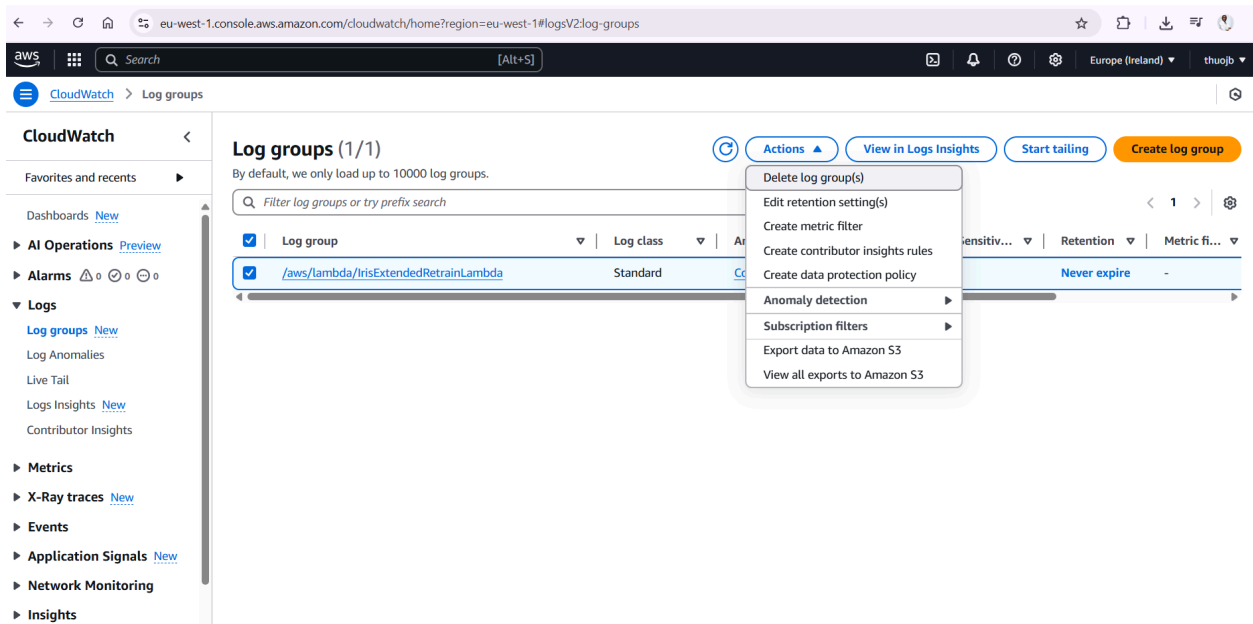
To confirm deletion, type "confirm" in the field.

confirm

Cancel Delete



Cloudwatch logs



eu-west-1.console.aws.amazon.com/cloudwatch/home?region=eu-west-1#alarmsV2?~:(alarmStateFilter~'ALARM)

aws

Search

[Alt+S]



Europe (Ireland)

thuojb

CloudWatch

Alarms

Favorites and recents

Dashboards [New](#)

▶ AI Operations [Preview](#)

▼ Alarms    

[In alarm](#)

All alarms

▼ Logs

Log groups [New](#)

Log Anomalies

Live Tail

Logs Insights [New](#)

Contributor Insights

▶ Metrics

▶ X-Ray traces [New](#)

▶ Events

▶ Application Signals [New](#)

▶ Network Monitoring

Alarms (0)

☐ Hide Auto Scaling alarms

Clear selection



Create composite alarm

Actions

Create alarm

Search

Alarm state: In alarm

Alarm type: Any

Actions status: Any

< 1 >



☐	Name	State	Last state update (UTC)	Conditions	Actions
No alarms No alarms to display Read more about Alarms Create alarm					

CloudShell

Feedback

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